

1) Oxidation number of Mg.
+2

Rule:- Mg is a group 2 metal,
so it normally forms a +2
by losing its 2 outer electrons.

~~Q. Physics:-~~

MCQs

1. C

2. A

3. D

4. C

5. A

6. A

7. D

8. B

9. C

10. D

11. A

12. A

13. C

14. A

15. D

16. B

17. A

18. C

19. B

20. D

21. D

22. C

23. C

24. D

Q: 2(i)

A: 2(i)

(a) least count = smallest division
main scale

$$= \frac{1}{50} = 0.02$$

$$0.02 \div 10 = 0.002 \text{ cm}$$

(b)

$$\text{m.s.r} = 6.7 \text{ cm}$$

best coincid. = 25th

$$\text{V.S.R} = 25 \times 0.002$$
$$0.05 \text{ cm}$$

$$\text{length} = \text{msr} + \text{VSR}$$
$$6.7 + 0.05$$
$$6.75$$

Q: 2(ii)

A: 2(ii)

Centre of mass

The point where the whole mass of body can be considered to be concentrated.

Centre of gravity

The point where the whole weight of body can be considered to be act.

(b) They aren't the same when gravitational field is non-uniform (it's strength or direction varies across the body).

In a uniform gravitational field, the centre of mass and centre of gravity are the same point.

Q. 2 (iii)

A. 2 (iii)

$$P = \rho gh$$

$$\rho_1 gh_1 = \rho_2 gh_2$$

$$13600 \times g \times 76 = 3400 \times g \times h$$
$$13600 \times 76 = 3400h$$

$$h = \frac{13600 \times 76}{3400}$$

$$h = 4 \times 76 = 304 \text{ cm.}$$

(b) for same pressure

$$h \propto \frac{1}{\rho}$$

Q: 3(i)
A: 3(i)

- (a) AB :- acceleration
BC :- zero acceleration rest/constant
CD :- deceleration

(b) Triangle :- $\frac{1}{2}(bh)$

$$\frac{1}{2}(10 \times 14)$$

$$+ \frac{70}{2}$$

$$70 \text{ m/s}^2$$

(c) distance travelled:-

$$\frac{1}{2}bh + L \times b + \frac{1}{2}bh$$

$$\frac{1}{2}(10 \times 14) + (20 \times 14) + \frac{1}{2}(10 \times 14)$$

$$70 + 280 + 70$$
$$= 420 \text{ m}$$

Q: 3(ii)

A: 3(ii)

(a)

* Renewable :-

Energy resources that are naturally replenished and can be used repeatedly.

Example:- Solar energy, wind energy

* Non-renewable :-

Energy resources that are finite and will eventually run out.

Example:- Coal, natural gas

(b) Hydroelectrical power station:-

Advantage:-

It produces electricity without burning fuel, so there is very little air pollution during operation.

Disadvantage:-

Building dams can flood large areas of land and affect ecosystems.

Conveyor belt.

Given :-

$$F = 350 \text{ N}$$

$$v = 2 \text{ m/s}$$

$$t = 90$$

i) work done

first find distance :-

$$d = vt$$

$$= 2 \times 90 = 180$$

work done :- $W = Fd$

$$350 \times 180$$

$$63000 \text{ J}$$

ii) Power output :-

$$P = \frac{W}{t}$$

$$= \frac{63000}{90}$$

$$= 700 \text{ W}$$